FVS-ACD v12.11: the maturity-dependent bias, addressed

Five gated levers, byte-identical gate-off, all four guardrails pass

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Wins on managed and mature stands. Developing Maine FIA stays open.

CONTEXT

v12.11 targets the one bias that flips sign with maturity

2018 ECANUSA: OSM edged FVS-ACD on merchantable volume increment
2026: v12.10.1 ships, one bias survives

1. **Site productivity: ESI holds**
2. **Data harmonization: one NSVB currency**
3. **New calibration: five gated levers**
4. **Integrating climate: 2030 ESI scenario**
5. **AI and productivity: caught our error**

Five agenda topics, what we bring

CONTEXT

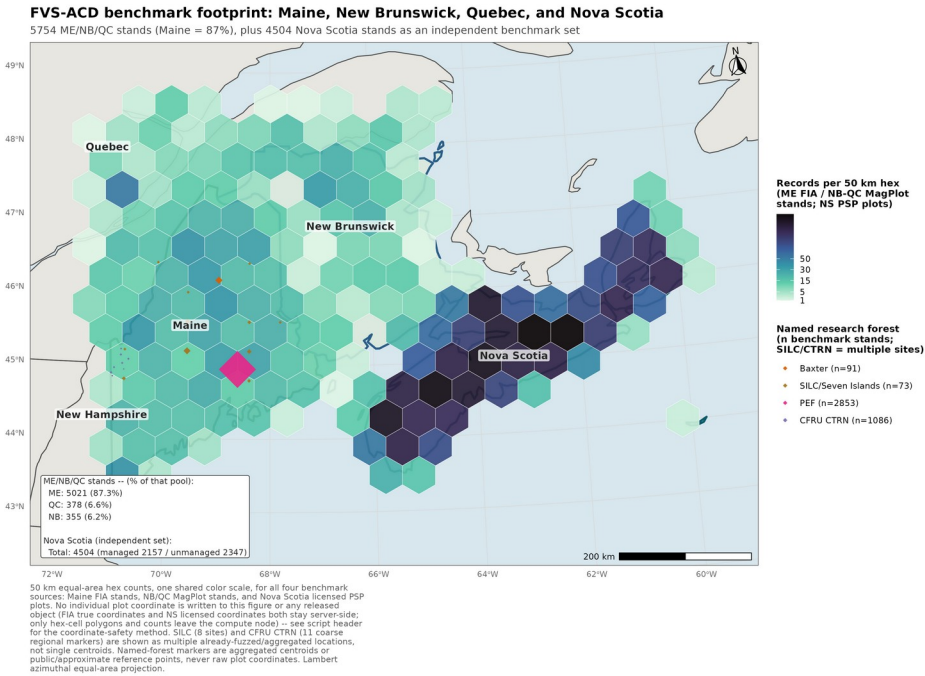
Six releases since the 2018 ECANUSA baseline, one bias at a time

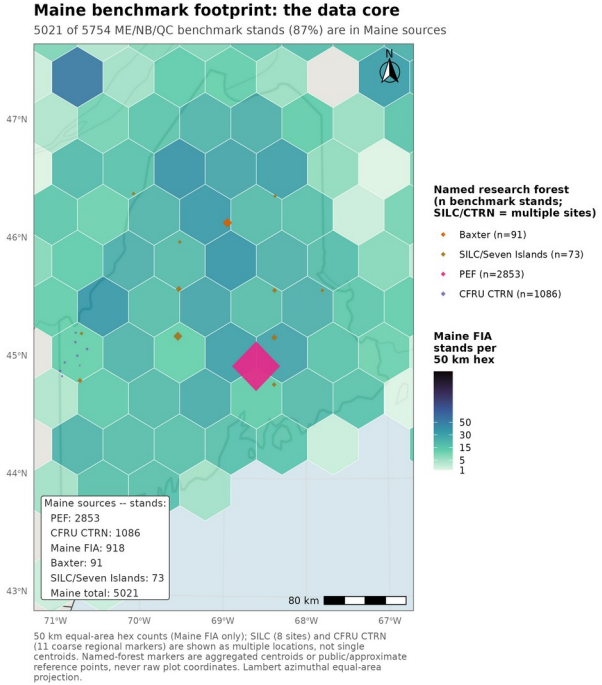
Release	What changed	Headline number
2018 to v12.7.1	Fixed the recruitment gate defect (0.95 cutoff suppressing ingrowth); coupled ingrowth to density	TOST 4/9 dips to 2/9, then recovers
v12.9.x	Gated SDImax correction, Quebec hardwood self-thinning edge	TOST recovers to 7/9, bias 4.9
v12.10.1	Chapter 11 management modifiers, gated releases on planted/PCT	Managed bias -19.0% to -2.5%, beats OSM
v12.11 (2026)	Five gated levers, maturity-dependent long-horizon bias	Beats OSM managed/mature; gate off 0.000e+00

2018 to 2026: the gap narrowed release by release, each refinement gated and individually isolated, reproducing its predecessor bit for bit when off. Gate off reproduces v12.10.1 bit for bit (max abs diff 0.000e+00), reconfirmed after every red-team fix.

DATA FOOTPRINT

Maine supplies 87.3% of the benchmark, Nova Scotia is independent





5,021 / 5,754
Maine stands, 87.3% of the pooled ME/NB/QC total

355
New Brunswick stands

378
Quebec stands

4,504
Nova Scotia stands, managed and unmanaged (independent test)

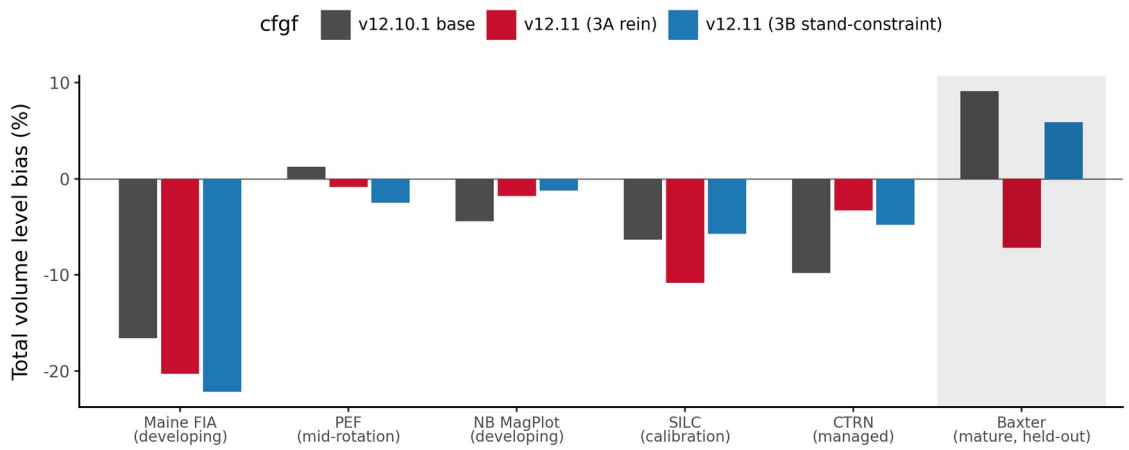
8
sources across four jurisdictions

Nova Scotia never used to fit or tune any lever Eight sources, one NSVB currency: PEF 2,853, CTRN 1,086, Maine FIA 918, Baxter 91, SILC 73, NB 355, QC 378.

THE PROBLEM

Volume bias flips sign across the maturity gradient

Total volume bias



Basal area bias

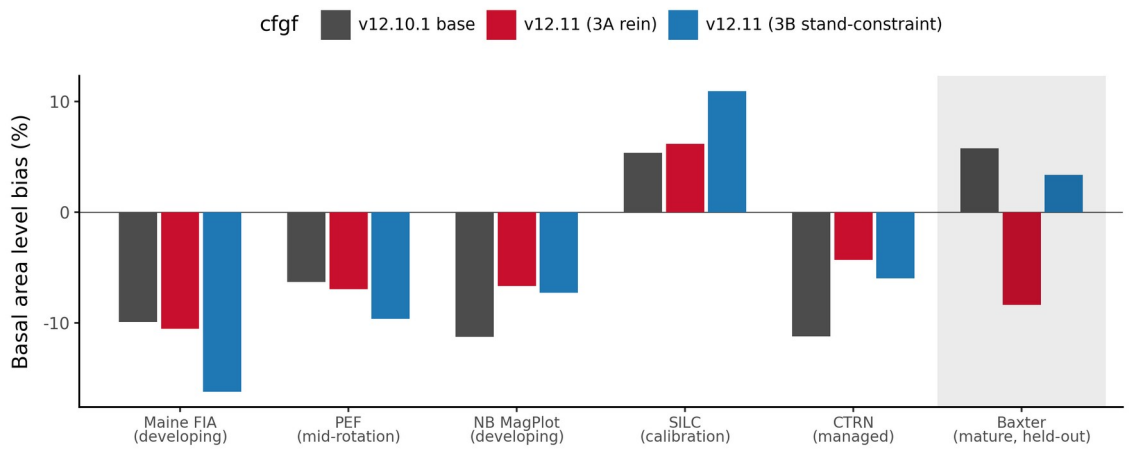


Figure 1. Total volume and basal area level bias across the stand-maturity gradient, base v12.10.1 versus the v12.11 candidate.

Base to v12.11 candidate, versus OSM v2.26.1
Maine FIA developing -16.60% to -16.13% (OSM -6.29%)
CTRN managed -9.78% to -2.24% (OSM -4.81%)
Baxter mature, held out +9.12% to +6.92% (OSM +12.37%)

THE FIX

Five gated levers, each off by default

1	**Recruitment**	IGMULT = 2.5, pooled ingrowth deficit
2	**Management-gated mortality**	fmin = 0.13, k = 12.4, RD0 = 0.38
3	**Chapter 10 stand constraint**	QMD floor 22 cm, binds only in mature stands
4	**Hardwood height by risk**	High-risk trees 10 to 11% shorter, 635 trees
5	**Crown ratio, investigated**	Already active, Russell et al. 2014

VALIDATION

All four guardrails pass, two flags remain open

Guardrail	Base v12.10.1	v12.11 candidate
1. Frozen in-domain FIA NSVB (n=733)	BA -11.28%, TOPHT +4.21%, TOTVOL -4.41%	BA -7.31%, TOPHT +5.88%, TOTVOL -1.50%
2. Bakuzis five relations (Reineke slope)	-1.59, PASS	-1.62 primary PASS; -1.88 managed sensitivity, FLAG only
3. Long-horizon plateau (~160 yr)	BA 36.10 m^2 ha^{-1}, plateaus	BA 35.29 m^2 ha^{-1}, plateaus
4. Managed out-of-sample vs OSM v2.26.1	Trails OSM on all 4 cells	Beats/ties OSM on 3 of 4: Baxter, CTRN, NB win; SILC basal area open

TOPHT drift traces to the per-province dht.mult scalar, not the new levers
Gate off reproduces v12.10.1 bit for bit (max abs diff 0.000e+00), reconfirmed after every red-team fix.

CENTRAL RESULT

On its target managed and mature sources, v12.11 beats OSM v2.26.1

Source	n	Total volume bias base / v12.11 / OSM	Basal area bias base / v12.11 / OSM
Baxter, ME (mature, held out)	91	+9.12% / +6.92% / +12.37%	+5.74% / +4.47% / +13.28%
CTRN, ME (managed)	1,086 / 1,033	-9.78% / -2.24% / -4.81%	-11.26% / -3.52% / -4.19%
NB MagPlot (managed)	355	-11.84% / -5.96% / -8.07%	-8.84% / -2.33% / -2.48%

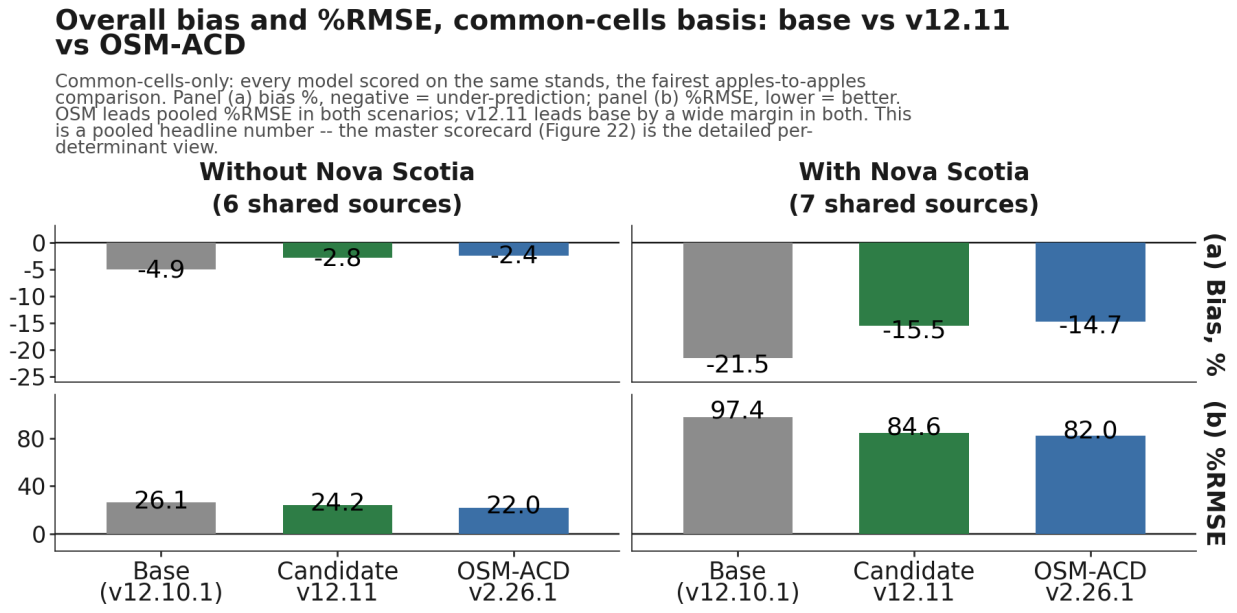
Two managed flags open: QC MagPlot volume, SILC basal area

All three sources scored at stand grain. QC MagPlot has no scored OSM comparison. OSM still leads developing Maine FIA at either grain (subplot gap moves only 0.08 points) -- the acknowledged v13.1 target.

OVERALL SUMMARY

Pooled headline, common cells: v12.11 narrows the gap to OSM

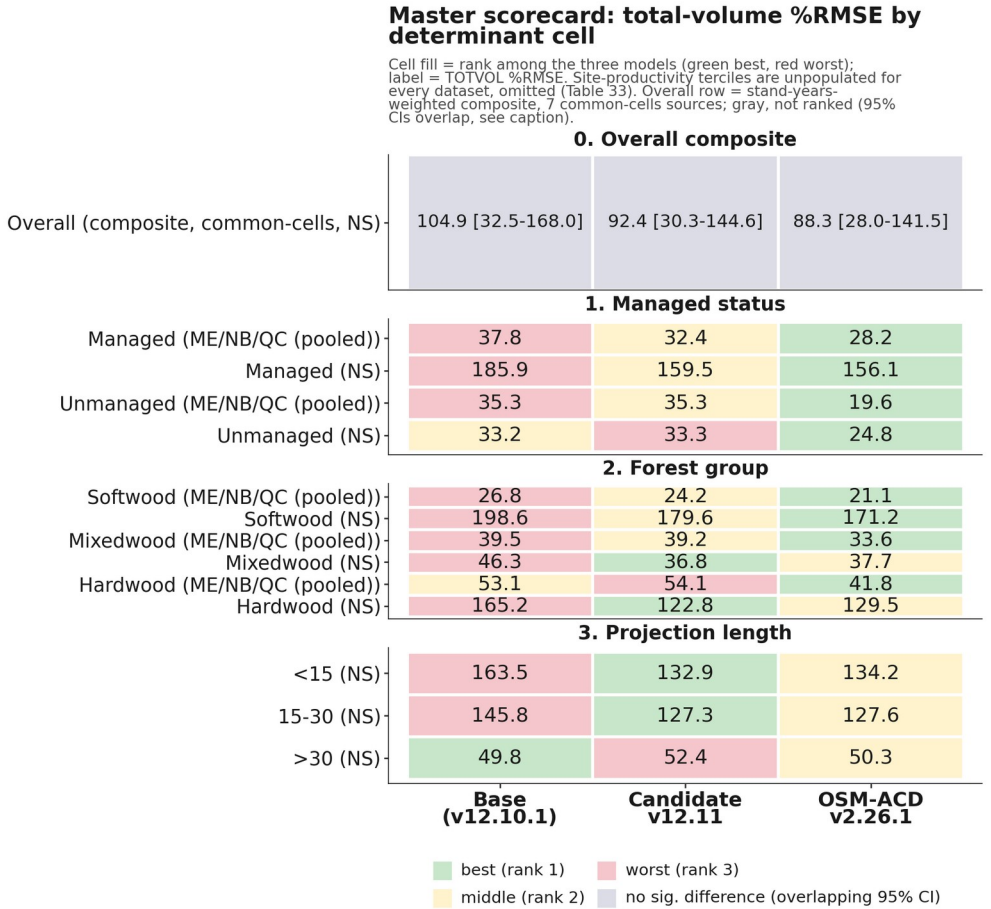
Without Nova Scotia (6 shared sources): bias -2.8% vs OSM -2.4%; %RMSE 24.2% vs OSM 22.0%. With Nova Scotia (7 shared sources): bias -15.5% vs OSM -14.7%; %RMSE 84.6% vs OSM 82.0%.



Common cells only, every model scored on the same stands. v12.11 leads base by a wide margin in both scenarios; OSM leads v12.11 by a narrower, consistent margin pooled. The master scorecard (next) is the detailed per-determinant view.

MASTER SCORECARD

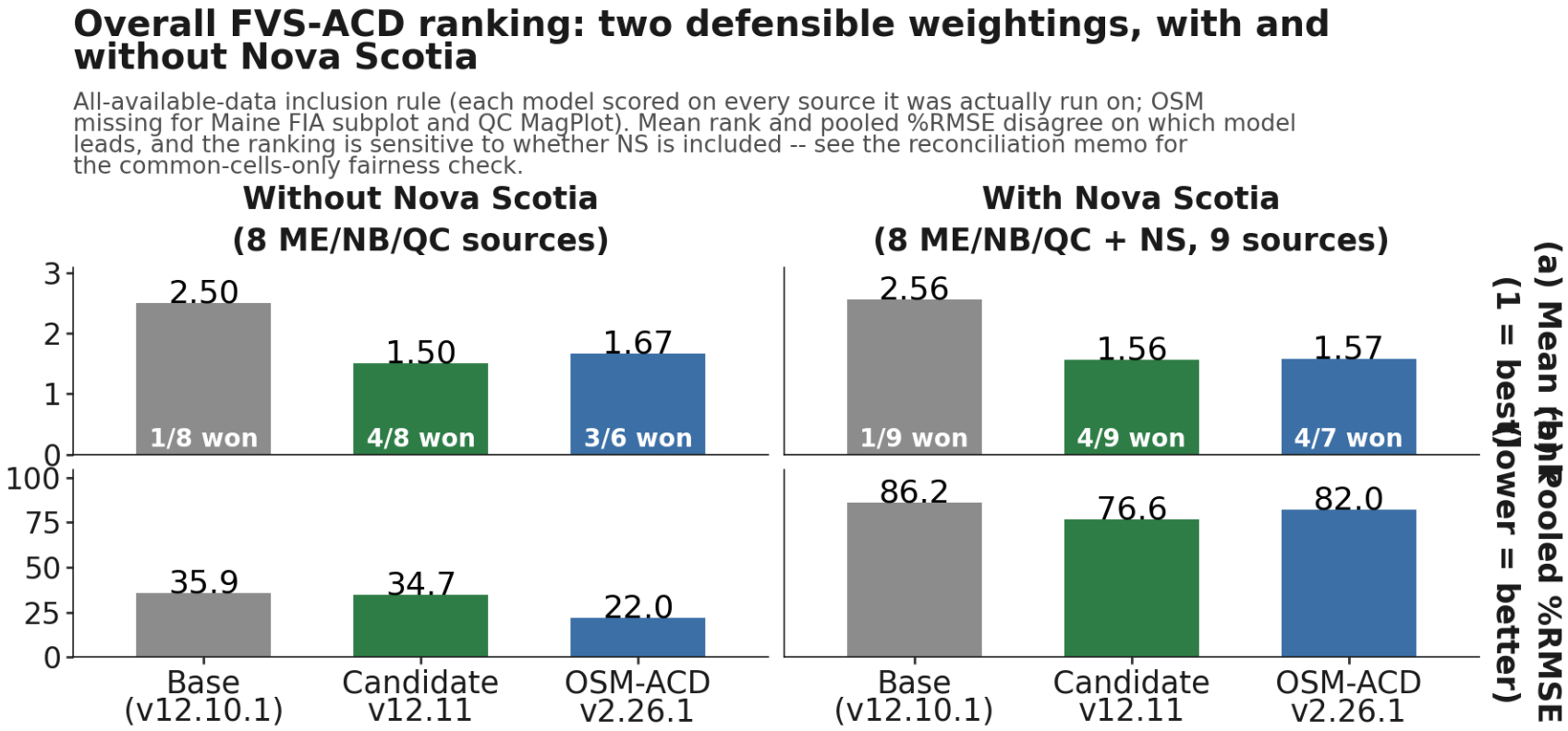
OSM leads managed; v12.11 wins Nova Scotia hardwood and mixedwood



Overall row = stand-years-weighted composite TOTVOL %RMSE (NS treatment; from figures/_data/composite_score_weighted.csv), 7 common-cells sources scored by all three models -- excludes Maine FIA subplot and QC MagPlot (OSM never run there) and the memat/medev pair (neither v12.11 nor OSM run there). Brackets = 95% bootstrap CI, source-level cluster resample, 7 clusters, n=2,000, seed 20260811. v12.11 and OSM overlap heavily there (equivalence, not a win) -- gray fill, not rank-colored. NS-long alternate (not plotted): Base 119.9 [32.5-149.6], Candidate 100.8 [30.3-123.9], OSM-ACD 100.2 [28.0-124.8]. Projection length, ME/NB/QC (pooled) is a structural n=0 gap (no source has a long enough span), not an unscored choice; see 20260811_final_scorecard_all_datasets.md Table 33. Data: master_scorecard.csv, composite_score_weighted.csv.

OVERALL RANKING (RECONCILED)

Neither model is a clean overall winner



n (sources): without NS 8 ME/NB/QC sources (n_total v12.11/base=8612, OSM=5319 stands); with NS 9 sources (n_total v12.11/base=13116, OSM=9818 stands; OSM never scored on Maine FIA subplot or QC MagPlot). Data: figures/_data/overall_ranking.csv. This is the all-available-data view; the common-cells-only fairness check (same stands for all three models) is in the reconciliation memo, 20260723_master_scorecard_and_ranking.md.

v12.11 more consistent, OSM more decisive where it wins

OSM lower pooled %RMSE under the common cells rule: without NS, OSM 22.0 vs v12.11 24.2; with NS, OSM 82.0 vs v12.11 84.6 (Table 34)

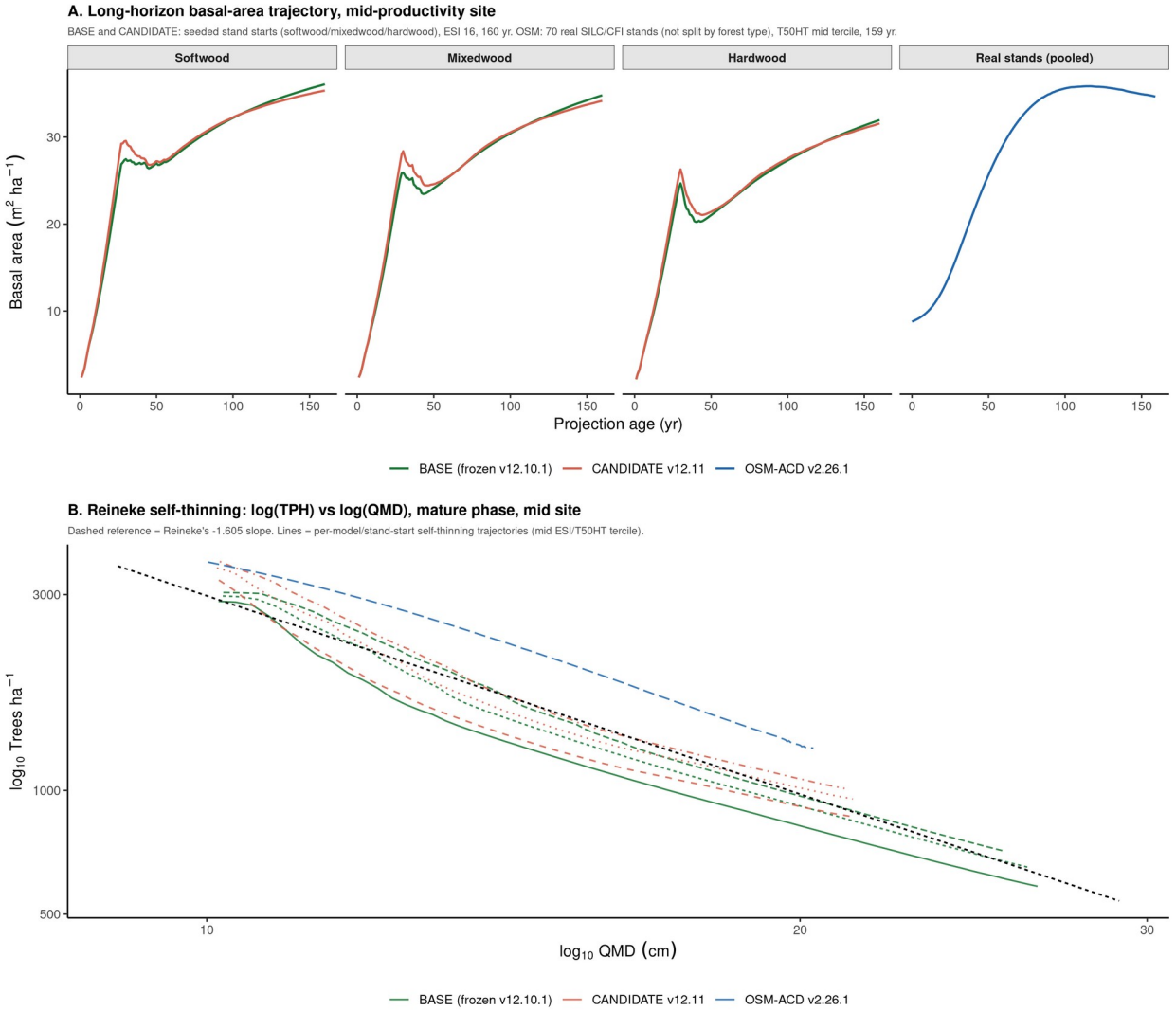
LONG-HORIZON BIOLOGICAL REALISM

v12.11 passes four of five Bakuzis relations over 160 years

Eichhorn: v12.11 pass, OSM flag, CV 26.3%

Site ordering: v12.11 a brief crossing, OSM a persistent inversion

OSM passes 3 of 5 (fails Eichhorn, CV 26.3%, and site ordering).



NOVA SCOTIA

The managed stand levers generalize to an independent province

1

4,504 scored pairs, fully independent

Nova Scotia PSP network, established 1965
Never used to fit v12.11

2

Managed total volume bias -58.10% to -42.32%

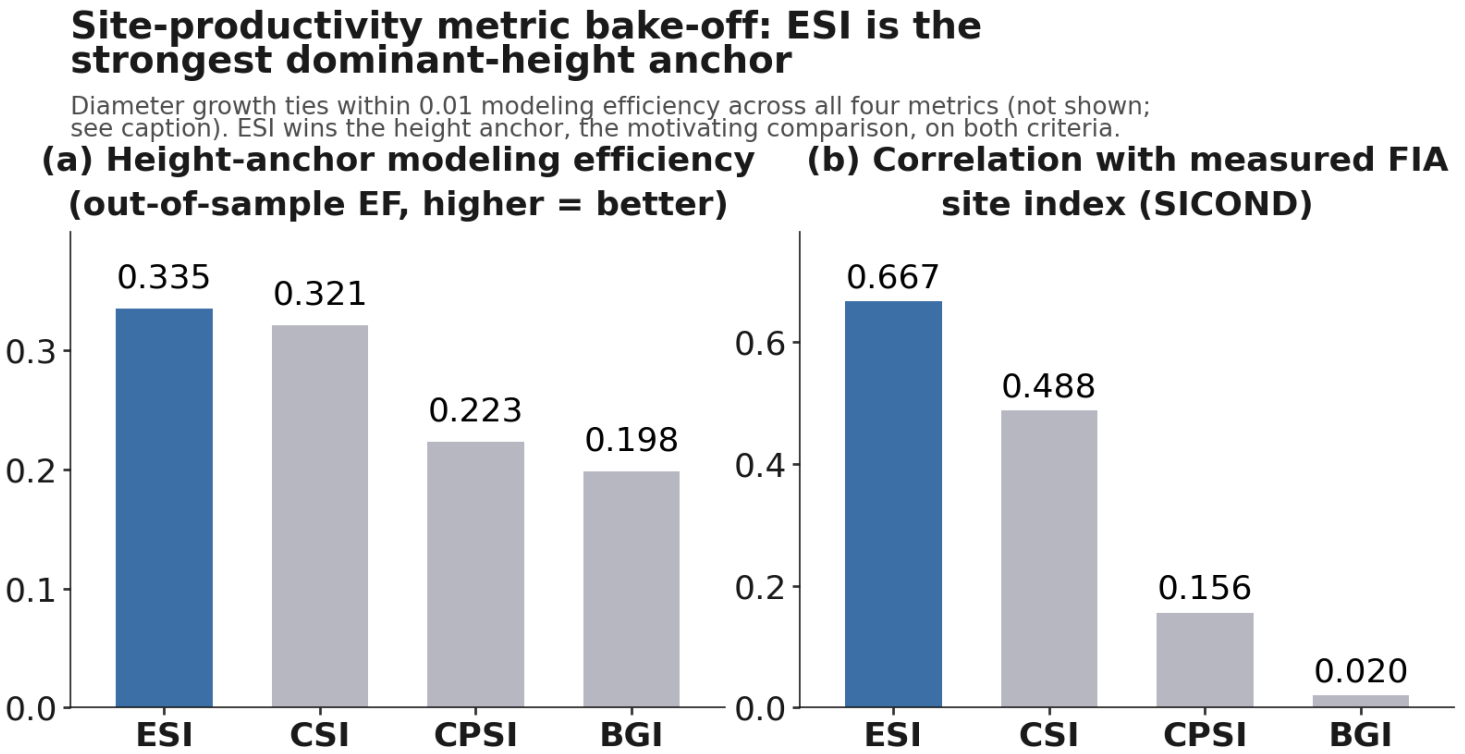
A 27% relative reduction

3

OSM head to head, and the integrity check

Managed: essentially tied (v12.11 -42.32% vs OSM -41.96%). Long horizon (>30 yr) is the clearest separation: v12.11 -1.66% vs OSM -9.14%. Site index confirmed on the same ESI basis as the model's anchor; a cut-detection bug (string vs integer) was caught and fixed before shipping.

Site productivity: keep ESI



Out-of-sample, 10-fold spatially blocked CV (folds by 0.25-degree cell); n = 15,346 plot-intervals (15,079 with all four metrics and clean anchor inputs; ESI extracted at 15,329/15,346 plots), Kuehne remeasurement pairs restricted to the ESI-covered US portion of the Acadian domain (ME/NH/VT). Uncertainty exception: this is a single out-of-sample point estimate per fold-averaged metric; per-fold spread (SD across the 10 folds) is not archived in the source memo, so no error bar is shown here -- stated explicitly rather than fabricated. Diameter growth: ESI 0.340, BGI 0.341, CPSI 0.348, CSI 0.338 EF, all within 0.01 of each other and not plotted. Source: 20260702_site_variable_bakeoff.md, Results 1-2.

Bake-off settled: ESI, BGI, CPSI, legacy CSI
Out of sample, spatially blocked CV
ESI wins the height anchor, ties on diameter growth
BGI and CPSI: no height signal, no cross-border coverage
Anchor correction tested, not shipped: +4% height offsets -10% basal area
Joint re-anchor is v13.0
Climate: ESI v8 current + 2030 surface, RCP4.5 +1.9 m (2030s) to +4.3 m (2090s) -- validated on observed climate only, never benchmarked on the projected surface.

ROADMAP

v13 is the fall arc: height, mortality, and the ask

1	**v13.0 Height and domain**	Gate off predictions outside the ESI range; blend sweep parked
2	**v13.1 Shipping decision**	Ships: ingrowth 2.5x plus r2 refit; SDImax retained, retuned onset 0.55-0.60. Two configurations blocked on biological realism: the full 3.9x ingrowth stack (Bakuzis gain reversal) and SDImax removed (fails on level).
3	**The ask of the group**	Provincial managed remeasurements, PSP treatment histories, and site surfaces

AI AND PRODUCTIVITY

AI accelerated the work, humans kept it honest

Accelerated	Weeks of benchmarking and figures compressed into days
Kept honest	A red team pass forced three reframes before v12.11 shipped
Reproducible	Every number traces to a script; gate off reproduces v12.10.1 bit for bit

VERDICT AND KEY TAKEAWAYS

v12.11 is ready, and its open gap is named

1 Gate off reproduces v12.10.1 bit for bit, 0.000e+00

2 Managed CTRN, NB MagPlot, and mature Baxter: v12.11 halves bias (Baxter +6.92 vs OSM +12.37; CTRN -2.24 vs OSM -4.81; NB -5.96 vs OSM -8.07)

3 Developing Maine FIA open: subplot grain moves -24.90% to -24.82%, a 0.08 point change

4 Nova Scotia bias weakens, never flips positive (managed long horizon: v12.11 -1.66% vs OSM -9.14%)

5 v13.1 ships ingrowth 2.5x plus r2, SDImax retained and retuned

Questions / Comments

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AcadianGY is open R, v12.6.1 to v12.11 now git tagged; plot coordinates stay on the analysis cluster, aggregates only.

Support: CRSF and CFRU, University of Maine | Maine-FOREST (NSF EPSCoR OIA-1849227) | PERSEUS (USDA NIFA)